

NANYANG JUNIOR COLLEGE

2008 JC2 PRELIMINARY EXAMINATION

MATHEMATICS

Higher 2

9740/01

8th September 2008

Paper 1

3 Hours

Additional Materials: Answer Paper

List of Formulae (MF 15)

READ THESE INSTRUCTIONS FIRST

Write your name, class and name of subject tutors in the space provided, as well as on all the work you hand in.

Write in dark blue or black pen on both sides of the paper.

You may use a soft pencil for any diagrams or graphs.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer all the questions.

Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.

You are expected to use a graphic calculator.

Unsupported answers from a graphic calculator are allowed unless a question specifically states otherwise.

Where unsupported answers from a graphic calculator are not allowed in a question, you are required to present the mathematical steps using mathematical notations and not calculator commands.

You are reminded of the need for clear presentation in your answers.

The number of marks is given in brackets [] at the end of each question or part question.

At the end of the examination, fasten all your work securely together.

NAME: _____

CLASS: _____ **SUBJECT TUTOR:** _____

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1		5		9	
2		6		10	
3		7		11	
4		8		TOTAL	

Answer All Questions

- 1 Sketch the graphs of $y = |x - 20|$ and $y = \frac{1}{x}$ on the same diagram, and hence or otherwise, solve the inequality $|x - 20| < \frac{1}{x}$. Leave your answer correct to 2 decimal places. [5]

- 2 (i) Find the expansion of $\frac{1}{(1 - 3x)^3}$ in ascending powers of x up to and including the x^2 term. [2]
 (ii) State the set of values of x for which the series expansion is valid. [1]
 (iii) Find, in terms of n , the coefficient of x^n in the expansion in ascending powers of x for $\frac{1}{(1 - 3x)^3}$, leaving your answer in the simplest form. [3]

- 3 A cubic curve with equation $f(x) = ax^3 + bx^2 + cx + d$ passes through the point $(0, 5)$ and has a stationary point at $(1, 2)$. The curve is above the x -axis for $0 \leq x \leq 1$. The area between the curve, the positive axes and the line $x = 1$ is $\frac{19}{6}$ units². Find the values of a, b, c and d . [4]
 Solve the equation $f(x) = \frac{1}{x}$, giving your answers to 3 decimal places. [2]

- 4 A geometric sequence $\{a_n\}$ has first term a and common ratio r . The sequence of numbers $\{b_n\}$ satisfy the relation $b_n = \ln(a_n)$ for $n \in \mathbb{N}^+$.
 (i) Show that $\{b_n\}$ is an arithmetic sequence and determine the value of the common difference in terms of r . [3]
 (ii) Find an expression for $\sum_{n=1}^{N+1} b_n$ in terms of a, a_{N+1} and N . [2]
 (iii) Hence, obtain an expression for $a_1 \times a_2 \times \dots \times a_{N+1}$ in terms of a, a_{N+1} and N . [2]

- 5 Let $U_r = \frac{1}{(r+1)(r+2)(r+3) \cdots (r+p)}$ where p is a positive integer.

Simplify $U_r - U_{r-1}$ as far as possible. [2]

Hence prove that $\sum_{r=1}^n \frac{1}{r(r+1)(r+2) \cdots (r+p)} = \frac{1}{p} \left[\frac{1}{p!} - \frac{1}{(n+1)(n+2) \cdots (n+p)} \right]$. [4]

Deduce the exact value of $\sum_{r=0}^{\infty} \frac{1}{(r+1)(r+2)(r+3)(r+4)}$. [2]

- 6 A sequence of numbers $\{x_n\}$ is defined by the relation $x_n - x_{n-1} = n - 1$ for all $n = 2, 3, 4, \dots$ and $x_1 = 1$.

(i) Write down the values of $\frac{2(x_n - 1)}{n - 1}$ for $n = 2, 3, 4$ and 5 . [2]

(ii) Hence make a conjecture for x_n in terms of n . [1]

(iii) Prove your conjecture in (ii) by induction. [5]

- 7 Consider the curve $y = \frac{(x-1)^2}{(x+3)(x-2)}$.

(i) State the coordinates of any point(s) of intersection with the axes. [1]

(ii) State the equations of the asymptotes. [2]

(iii) On separate diagrams, sketch each of the following curves

(a) $y = \frac{(x-1)^2}{(x+3)(x-2)}$ [3]

(b) $y^2 = \frac{(x-1)^2}{(x+3)(x-2)}$ [2]

(c) $y = \frac{x^2 - 2|x| + 1}{(|x| + 3)(|x| - 2)}$ [2]

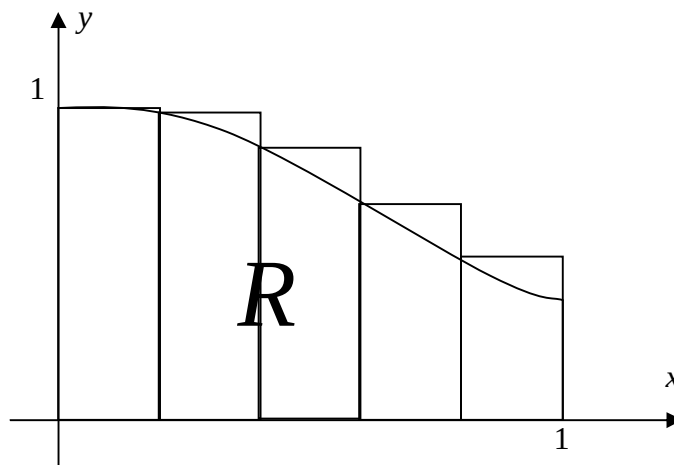
indicating clearly the intercept(s), asymptote(s) & coordinates of the stationary point(s) of each curve.

- 8 (a) A petrol tanker is damaged in a road accident, and petrol leaks onto a flat section of a motorway. The leaking petrol begins to spread in a circle of thickness 2 mm. Petrol is leaking from the tanker at a rate of $0.0084 \text{ m}^3\text{s}^{-1}$. Find the rate at which the radius of the circle of petrol is increasing at the instant when the radius of the circle is 3 m, giving your answer in m s^{-1} to 2 decimal places. [4]

- (b) A curve is defined parametrically by $x = \frac{2t}{t+1}$, $y = \frac{t^2}{t+1}$. Find the equation of the normal to the curve at the point $P\left(1, \frac{1}{2}\right)$. The normal at P meets the curve again at Q . Find the exact coordinates of Q . [8]

- 9 (a) Prove that $\frac{d}{dx}(x \sin x + \cos x) = x \cos x$ and hence find the exact value of $\int_0^{\pi} x |\cos x| dx$. [3]

(b)



The diagram (not to scale) shows the region R bounded by the axes, the curve $y = (x^2 + 1)^{-\frac{3}{2}}$ and the line $x = 1$. The integral $\int_0^1 (x^2 + 1)^{-\frac{3}{2}} dx$ is denoted by I .

- (i) By calculating the area of the 5 rectangles of equal width as shown in the diagram above, estimate the area of R and state whether it is an underestimate or overestimate of the actual area of R . [3]
- (ii) Use the substitution $x = \tan \theta$ to show that $I = \frac{1}{2}\sqrt{2}$. [4]
- (iii) Determine the volume of the solid generated when R is rotated 2π radians about the x -axis. [2]

- 10 (a)** In the Argand diagram, the point P represents the complex number z . Draw a clearly labelled diagram to show the locus of P when z satisfies

(i) $|z - 3 - 4i| = 5$ [1]

(ii) $|z - 2 + 3i| = |z - 10 - 3i|$ [2]

Hence find the greatest and least possible values of $|z|$ when $|z - 3 - 4i| \leq 5$ and $|z - 2 + 3i| \geq |z - 10 - 3i|$. [3]

- (b)** Find in the form $re^{i\theta}$, where $-\pi < \theta \leq \pi$, all complex numbers z satisfying the equation

$$z^6 = \frac{5+i}{2+3i}. \quad [3]$$

- (c)** If $z = e^{i\theta}$, where $z \neq 1$, show that

$$\left| 1 + z + z^2 + \dots + z^{n-1} \right| = \left| \frac{\sin\left(\frac{1}{2}n\theta\right)}{\sin\left(\frac{1}{2}\theta\right)} \right|. \quad [3]$$

- 11** Two planes Π_1 and Π_2 have equations $\mathbf{r} = (3 + \lambda - 2\mu)\mathbf{i} + (2 - 3\lambda + \mu)\mathbf{j} + (1 - \mu)\mathbf{k}$ and $\mathbf{r} = \begin{pmatrix} a \\ -1 \\ b \end{pmatrix} + s\mathbf{i} + t\mathbf{j} + u\mathbf{k}$ respectively.

respectively.

- (i) Show that the Cartesian equation of the plane Π_1 is $3x + y - 5z = 6$. [2]
- (ii) Given that the point $A(3, 2, 1)$ lies on Π_2 and that the two planes are perpendicular to each other, find the values of a and of b . [3]
- (iii) Find a vector equation of l , the line of intersection between the planes Π_1 and Π_2 . [2]
- (iv) B is a point with position vector $-6\mathbf{i} + 3\mathbf{j} + 11\mathbf{k}$. Find the coordinates of the point C on Π_2 such that $\angle ACB = 90^\circ$. [3]
- (v) The line L is the reflection of the line AB in the plane Π_2 . Find an equation for the line L and determine the acute angle between the lines l and L . [4]

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